1. Student Roll Number Search

Problem Statement:
A teacher has a list of student roll numbers stored in the order they were registered. The teacher wants to check whether a particular roll number belongs to a student in the class.

Test Case

Input:

Roll Numbers: [101, 105, 110, 115, 120]
Search Roll Number: 110

Output:

Found

---

2. Finding a Book on a Shelf

Problem Statement:
A librarian maintains a list of books arranged by their accession number. A student asks for a specific book. Determine the position of the book in the list.

Test Case

Input:

Book IDs: [501, 503, 507, 510, 515]
Book ID to Search: 510

Output:

3

---

3. Attendance Verification

Problem Statement:
A class attendance sheet contains student IDs. The teacher wants to know how many times a particular ID appears due to duplicate entries.

Test Case

Input:

Student IDs: [12, 15, 12, 18, 12, 20]
Target ID: 12

Output:

3

---

4. Highest Exam Score

Problem Statement:
A school has recorded marks of students. Find the highest score obtained in the examination.

Test Case

Input:

Marks: [78, 92, 65, 88, 95]

Output:

95

---

5. Lowest Temperature Recorded

Problem Statement:
A weather station stores daily temperature readings. Find the lowest temperature recorded during the week.

Test Case

Input:

Temperatures: [34, 29, 31, 27, 35, 30]

Output:

27

---

6. Employee ID Lookup

Problem Statement:
A company maintains employee IDs in sorted order. HR wants to verify whether a given employee ID exists in the database.

Test Case

Input:

Employee IDs: [1001, 1005, 1010, 1015, 1020]
Search ID: 1010

Output:

Found

---

7. Product Search in Inventory

Problem Statement:
An online store keeps product IDs sorted in ascending order. A customer enters a product ID to check its availability. Determine its position in the inventory.

Test Case

Input:

Product IDs: [200, 250, 300, 350, 400]
Search ID: 300

Output:

2

---

8. First Customer Order

Problem Statement:
A sorted database contains multiple entries for customer orders. Find the position of the first order placed by a specific customer.

Test Case

Input:

Order IDs: [101, 105, 105, 105, 110, 115]
Target Order ID: 105

Output:

1

---

9. Last Train Booking Record

Problem Statement:
A railway booking system stores booking IDs in sorted order. Find the last occurrence of a booking ID.

Test Case

Input:

Booking IDs: [20, 25, 25, 25, 30, 35]
Target ID: 25

Output:

3

---

10. Popular Product Analysis

Problem Statement:
A store manager wants to know how many times a product appears in the sorted sales records.

Test Case

Input:

Sales Records: [5, 5, 5, 8, 10, 10]
Product ID: 5

Output:

3

---

11. Organizing Student Marks

Problem Statement:
A teacher wants to arrange student marks from lowest to highest before preparing the final report card.

Test Case

Input:

Marks: [78, 45, 92, 67, 85]

Output:

[45, 67, 78, 85, 92]

---

12. Ranking Participants

Problem Statement:
An event organizer wants to display participant scores from highest to lowest to prepare the leaderboard.

Test Case

Input:

Scores: [85, 92, 78, 95, 88]

Output:

[95, 92, 88, 85, 78]

---

13. Tracking Data Movement

Problem Statement:
A software engineer wants to know how many exchanges occur while arranging a list of numbers in ascending order.

Test Case

Input:

Numbers: [5, 1, 4, 2]

Output:

Sorted Numbers: [1, 2, 4, 5]
Total Swaps: 4

---

14. Selecting Scholarship Students

Problem Statement:
A college wants to arrange students based on their scores by repeatedly selecting the smallest score from the remaining unsorted list.

Test Case

Input:

Scores: [64, 25, 12, 22, 11]

Output:

[11, 12, 22, 25, 64]

---

15. Arranging Library Books

Problem Statement:
A librarian inserts returned books into their correct position on a shelf one by one. Arrange the books in ascending order.

Test Case

Input:

Book IDs: [12, 11, 13, 5, 6]

Output:

[5, 6, 11, 12, 13]

---

16. Seating Arrangement

Problem Statement:
Students arrive at different times and must be seated according to their roll numbers. Arrange all roll numbers in ascending order.

Test Case

Input:

Roll Numbers: [25, 10, 30, 15, 20]

Output:

[10, 15, 20, 25, 30]

---

17. Sorting Customer Orders

Problem Statement:
An e-commerce company receives customer order values in random order. Arrange them from smallest to largest.

Test Case

Input:

Order Values: [380, 270, 430, 30, 90, 820, 100]

Output:

[30, 90, 100, 270, 380, 430, 820]

---

18. Merging Branch Records

Problem Statement:
Two bank branches maintain customer account numbers in sorted order. Merge both records into a single sorted list.

Test Case

Input:

Branch A: [1, 3, 5]
Branch B: [2, 4, 6]

Output:

[1, 2, 3, 4, 5, 6]

---

19. Tournament Rankings

Problem Statement:
A sports committee receives player scores from different regions. Arrange all scores in ascending order before publishing the rankings.

Test Case

Input:

Scores: [38, 27, 43, 3, 9, 82, 10]

Output:

[3, 9, 10, 27, 38, 43, 82]

---

20. Hospital Patient Records

Problem Statement:
Two hospital departments maintain patient IDs separately in sorted order. Combine both lists into a single sorted patient record.

Test Case

Input:

Department A: [101, 103, 105]
Department B: [102, 104, 106]

Output:

[101, 102, 103, 104, 105, 106]